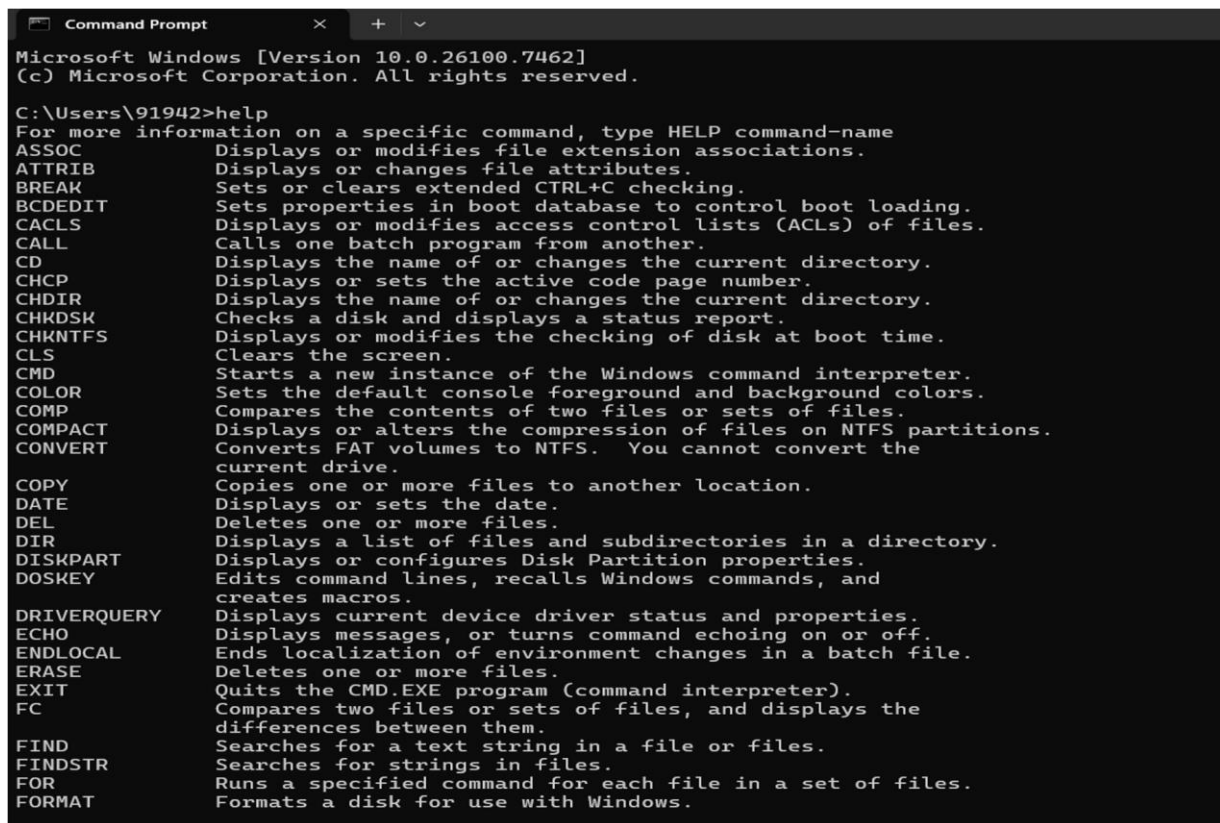


PRACTICAL 1

John has recently started a new job as system administrator in MNC. He is dealing with computers having different operating systems such as Windows, LINUX and UNIX. How will he be able to interact with the system by using different operating system commands for the following operation?

Que.1. Display the user manual of any command that we can run on the terminal.



```
Command Prompt
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\91942>help
For more information on a specific command, type HELP command-name
ASSOC          Displays or modifies file extension associations.
ATTRIB         Displays or changes file attributes.
BREAK          Sets or clears extended CTRL+C checking.
BCDEDIT        Sets properties in boot database to control boot loading.
CACLS          Displays or modifies access control lists (ACLs) of files.
CALL           Calls one batch program from another.
CD             Displays the name of or changes the current directory.
CHCP           Displays or sets the active code page number.
CHDIR          Displays the name of or changes the current directory.
CHKDSK         Checks a disk and displays a status report.
CHKNTFS        Displays or modifies the checking of disk at boot time.
CLS            Clears the screen.
CMD            Starts a new instance of the Windows command interpreter.
COLOR          Sets the default console foreground and background colors.
COMP           Compares the contents of two files or sets of files.
COMPACT        Displays or alters the compression of files on NTFS partitions.
CONVERT        Converts FAT volumes to NTFS. You cannot convert the
               current drive.
COPY           Copies one or more files to another location.
DATE           Displays or sets the date.
DEL            Deletes one or more files.
DIR            Displays a list of files and subdirectories in a directory.
DISKPART       Displays or configures Disk Partition properties.
DOSKEY         Edits command lines, recalls Windows commands, and
               creates macros.
DRIVERQUERY    Displays current device driver status and properties.
ECHO           Displays messages, or turns command echoing on or off.
ENDLOCAL       Ends localization of environment changes in a batch file.
ERASE          Deletes one or more files.
EXIT           Quits the CMD.EXE program (command interpreter).
FC             Compares two files or sets of files, and displays the
               differences between them.
FIND           Searches for a text string in a file or files.
FINDSTR        Searches for strings in files.
FOR            Runs a specified command for each file in a set of files.
FORMAT         Formats a disk for use with Windows.
```

```
Command Prompt
FOR          Runs a specified command for each file in a set of files.
FORMAT       Formats a disk for use with Windows.
FSUTIL       Displays or configures the file system properties.
FTYPE        Displays or modifies file types used in file extension
associations.
GOTO         Directs the Windows command interpreter to a labeled line in
a batch program.
GPRESULT     Displays Group Policy information for machine or user.
HELP         Provides Help information for Windows commands.
ICACLS       Display, modify, backup, or restore ACLs for files and
directories.
IF           Performs conditional processing in batch programs.
LABEL        Creates, changes, or deletes the volume label of a disk.
MD           Creates a directory.
MKDIR        Creates a directory.
MKLINK       Creates Symbolic Links and Hard Links
MODE         Configures a system device.
MORE         Displays output one screen at a time.
MOVE         Moves one or more files from one directory to another
directory.
OPENFILES    Displays files opened by remote users for a file share.
PATH         Displays or sets a search path for executable files.
PAUSE        Suspends processing of a batch file and displays a message.
POPD         Restores the previous value of the current directory saved by
PUSHD.
PRINT        Prints a text file.
PROMPT       Changes the Windows command prompt.
PUSHD        Saves the current directory then changes it.
RD           Removes a directory.
RECOVER      Recovers readable information from a bad or defective disk.
REM          Records comments (remarks) in batch files or CONFIG.SYS.
REN          Renames a file or files.
RENAME       Renames a file or files.
REPLACE      Replaces files.
RMDIR        Removes a directory.
ROBOCOPY     Advanced utility to copy files and directory trees
SET          Displays, sets, or removes Windows environment variables.
SETLOCAL     Begins localization of environment changes in a batch file.
SC           Displays or configures services (background processes).
SCHTASKS     Schedules commands and programs to run on a computer.
SHIFT        Shifts the position of replaceable parameters in batch files.
```

```
Command Prompt
PUSHD.
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SC           Displays or configures services (background processes).
SCHTASKS     Schedules commands and programs to run on a computer.
SHIFT        Shifts the position of replaceable parameters in batch files.
SHUTDOWN     Allows proper local or remote shutdown of machine.
SORT         Sorts input.
START        Starts a separate window to run a specified program or command.
SUBST        Associates a path with a drive letter.
SYSTEMINFO   Displays machine specific properties and configuration.
TASKLIST     Displays all currently running tasks including services.
TASKKILL     Kill or stop a running process or application.
TIME         Displays or sets the system time.
TITLE        Sets the window title for a CMD.EXE session.
TREE         Graphically displays the directory structure of a drive or
path.
TYPE         Displays the contents of a text file.
VER          Displays the Windows version.
VERIFY       Tells Windows whether to verify that your files are written
correctly to a disk.
VOL          Displays a disk volume label and serial number.
XCOPY        Copies files and directory trees.
WMIC         Displays WMI information inside interactive command shell.

For more information on tools see the command-line reference in the online help.
```

2. Change the current working directory of the user

```
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>cd OneDrive

C:\Users\HP\OneDrive>dir
Volume in drive C has no label.
Volume Serial Number is 280D-B4AF

Directory of C:\Users\HP\OneDrive

17-01-2026  21:21    <DIR>          .
17-01-2026  22:58    <DIR>          ..
29-03-2025  07:42    <DIR>          Attachments
28-03-2025  12:55                (0) Book 1.xlsx
28-03-2025  12:55                (0) Book 2.xlsx
28-03-2025  12:55                (0) Book.xlsx
19-01-2026  22:38    <DIR>          Desktop
12-01-2026  19:31    <DIR>          Documents
28-03-2025  12:47    (1,053,417) Getting started with OneDrive.pdf
17-01-2026  21:21                1,158 Personal Vault.lnk
12-01-2026  19:31    <DIR>          Pictures
                    5 File(s)      1,054,575 bytes
                    6 Dir(s)  127,678,513,152 bytes free

C:\Users\HP\OneDrive>cd Desktop
```

3. Lists all the contents in the current working directory

```
C:\Users\HP\OneDrive>cd Desktop

C:\Users\HP\OneDrive\Desktop>dir
Volume in drive C has no label.
Volume Serial Number is 280D-B4AF

Directory of C:\Users\HP\OneDrive\Desktop

19-01-2026  22:38    <DIR>          .
17-01-2026  21:21    <DIR>          ..
14-10-2023  16:48                2,057 Hotstar.lnk
08-01-2026  16:11    151,621 MHHSRP991AFE31-911A-4608012026160920011.pdf
08-01-2026  15:50    151,260 MHHSRPADAD8F08-1621-4008012026154807879.pdf
19-01-2026  21:27    <DIR>          Practical
18-12-2025  11:17    12,903 WhatsApp Image 2025-11-28 at 1.41.36 PM (1).jpeg
                    4 File(s)      317,841 bytes
                    3 Dir(s)  127,675,105,280 bytes free
```

4. Read, modify or concatenate text files • Read a text file:

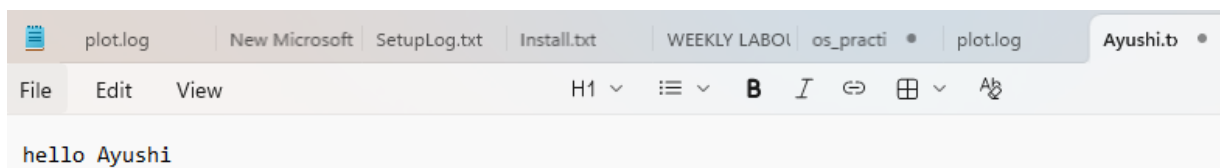
```
C:\Users\HP\OneDrive\Desktop>echo hello Ayushi > Ayushi.txt.txt

C:\Users\HP\OneDrive\Desktop>type Ayushi.txt.txt
hello Ayushi
```

- Modify a text file:

```
C:\Users\HP\OneDrive\Desktop>notepad Ayushi.txt.txt  
C:\Users\HP\OneDrive\Desktop>Ayushi.txt.txt Name Ayushis.txt
```

- Concatenate text files:



5. Create a new directory under any directory

```
C:\Users\HP\OneDrive\Desktop> mkdir Practical  
A subdirectory or file Practical already exists.  
  
C:\Users\HP\OneDrive\Desktop>md Practical  
A subdirectory or file Practical already exists.  
  
C:\Users\HP\OneDrive\Desktop>cd  
C:\Users\HP\OneDrive\Desktop
```

6. Display the current working directory of the terminal

```
C:\Users\HP\OneDrive\Desktop>cd  
C:\Users\HP\OneDrive\Desktop
```

7. Write its arguments to standard output

```
C:\Users\HP\OneDrive\Desktop>echo Practical OS  
Practical OS
```

8. Remove a file.

```
C:\Users\HP\OneDrive\Desktop>del A.txt  
C:\Users\HP\OneDrive\Desktop>|
```

9. Delete a directory

```
C:\Users\HP\OneDrive\Desktop>  
C:\Users\HP\OneDrive\Desktop>rmdir /s Practical  
Practical, Are you sure (Y/N)? yes  
C:\Users\HP\OneDrive\Desktop>|
```

10. Copy a file or directory

```
C:\Users\HP\OneDrive\Desktop>copy Ayushi.txt.txt Ayushi.txt  
1 file(s) copied.
```

11. Switch to root-user

```

C:\Users\HP\OneDrive\Desktop>runas /user:Administrator
RUNAS USAGE:

RUNAS [ [/noprofile | /profile] [/env] [/savecred | /netonly] ]
        /user:<UserName> program

RUNAS [ [/noprofile | /profile] [/env] [/savecred] ]
        /smartcard [/user:<UserName>] program

RUNAS [ [/machine:<MachineType>] ] /trustlevel:<TrustLevel> program

        /noprofile      specifies that the user's profile should not be loaded.
                        This causes the application to load more quickly, but
                        can cause some applications to malfunction.
        /profile        specifies that the user's profile should be loaded.
                        This is the default.
        /env            to use current environment instead of user's.
        /netonly        use if the credentials specified are for remote
                        access only.
        /savecred       to use credentials previously saved by the user.
        /smartcard      use if the credentials are to be supplied from a
                        smartcard.
        /user           <UserName> should be in form USER@DOMAIN or DOMAIN\USER
        /showtrustlevels displays the trust levels that can be used as arguments
                        to /trustlevel.
        /trustlevel     <Level> should be one of levels enumerated
                        in /showtrustlevels.
        /machine        specifies the machine architecture of the process.
                        <MachineType> should be one of x86|amd64|arm|arm64.
        program         command line for EXE. See below for examples

Examples:
> runas /noprofile /user:mymachine\administrator cmd
> runas /profile /env /user:mydomain\admin "mmc %windir%\system32\dsa.msc"
> runas /env /user:user@domain.microsoft.com "notepad \"my file.txt\""

NOTE: Enter user's password only when prompted.
NOTE: /profile is not compatible with /netonly.
NOTE: /savecred is not compatible with /smartcard.

```

12. Moves files and directories from one directory to another

```

C:\Users\HP\OneDrive\Desktop>move Ayushi.txt.txt D:\Downloads
The file exists.
        0 file(s) moved.

```

13. Search for a particular string/word in a text file

```

C:\Users\HP\OneDrive\Desktop>find "ayushi" Ayushi.txt

----- AYUSHI.TXT

```

14. Print the top N number of data of the given input

```
C:\Users\HP\OneDrive\Desktop>more Ayushi.txt
hello Ayushi
aa
bb
cc
dd
end
```

15. Print the last N number of data of the given input

```
C:\Users\HP\OneDrive\Desktop>type Ayushi.txt
hello Ayushi
aa
bb
cc
dd
end
```

16. Remove read permission from owner of the file

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /deny Everyone:R
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>|
```

17. Add read permission to owner plus re move write permission and add write permission to group

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Everyone:R
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>|
```

18. Add write permission to owner no permission to others

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Everyone:W
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files
```

19. Assign permission to users

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Users:F
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>|
```

20. Assign R/W/X to others

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Everyone:F
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>|
```

21. Remove all permission from all users

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /remove Everyone
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>|
```

22. Remove read permission from user using absolute mode

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /deny Everyone:R
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>|
```

23. Add R/w permission to owner no permission to group and group and other

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Everyone:F
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /remove Users
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /remove Everyone
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files
```


24. Add execute permission to owner, read to group and others

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Users:R
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files

C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Everyone:R
processed file: Ayushi.txt
Successfully processed 1 files; Failed processing 0 files
```

25. Add execute permission to all user

```
C:\Users\HP\OneDrive\Desktop>icacls Ayushi.txt /grant Everyone:X
```

IN LINUX

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (master)
$ git config --global user.name
[Ayushi godbole]

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (master)
$ git config --global user.email
ayushigodbole1@gmail.com

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (master)
$ dir
Ayushi.txt      MHHSRP991AFE31-911A-4608012026160920011.pdf  RAJNAGAR\ COMMERCIAL\ 2017.dwl2          ~$WEEKLY\ LABOUR\ PAYMENT.xlsx
Ayushi.txt.txt  MHHSRPADAD8F08-1621-4008012026154807879.pdf  whatsapp\ Image\ 2025-11-28\ at\ 1.41.36\ PM\ (1).jpeg  ~$ave\ of\ Application.docx
Hotstar.lnk     RAJNAGAR\ COMMERCIAL\ 2017.dwl                desktop.ini

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (master)
$ cd OS_CD24010
bash: cd: OS_CD24010: No such file or directory

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (master)
$ cd OS_CD24010

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ ls
Prelab.pdf

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git add Prelab.pdf

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   Prelab.pdf

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    ../Ayushi.txt
    ../Ayushi.txt.txt
    ../Hotstar.lnk
    ../MHHSRP991AFE31-911A-4608012026160920011.pdf
    ../MHHSRPADAD8F08-1621-4008012026154807879.pdf
```

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git commit -m"Prelab pdf executed successfully"
[master (root-commit) 0274831] Prelab pdf executed successfully
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 OS_CD24010/Prelab.pdf

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    ../Ayushi.txt
    ../Ayushi.txt.txt
    ../Hotstar.lnk
    ../MHHSRP991AFE31-911A-4608012026160920011.pdf
    ../MHHSRPADAD8F08-1621-4008012026154807879.pdf
    ../RAJNAGAR COMMERCIAL 2017.dwl
    ../RAJNAGAR COMMERCIAL 2017.dwl2
    ../WhatsApp Image 2025-11-28 at 1.41.36 PM (1).jpeg
    ../desktop.ini
    ../~$WEEKLY LABOUR PAYMENT.xlsx
    ../~$ave of Application.docx

nothing added to commit but untracked files present (use "git add" to track)

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git add -u

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git add practical1.pdf
fatal: pathspec 'practical1.pdf' did not match any files

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (master)
$ git branch -M main

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (main)
$ git remote add origin https://github.com/Ayushigodbole/OS-CD24010.git

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop/OS_CD24010 (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 1.46 MiB | 717.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```